

Functions in Java

```
{  
    int i = 4;  
    int j = 5;  
    int sum = i + j;  
    System.out.println (sum);  
    .....  
    int k = 9;  
    int l = 10;  
    ..... (k+l);  
}
```

} repeated code

function is a block of code which is reusable,
we can use it by calling the funcn in our code.

```
public static void main (String [] args) {  
    int i = 4;  
    int j = 5;  
    int result = sum (i, j);  
    System.out.println (result);  
    sum (10, 9);  
}
```

calling a funcn

arguments

returning an output

```
{  
    int Sum (int a, int b) {  
        int result = a + b;  
        return result;  
    }  
}
```

static

return type

parameters

name

return type — data type
i.e is written
by funcn

funcn syntax $\Rightarrow$ static datatype name (parameters) {
 // code
}

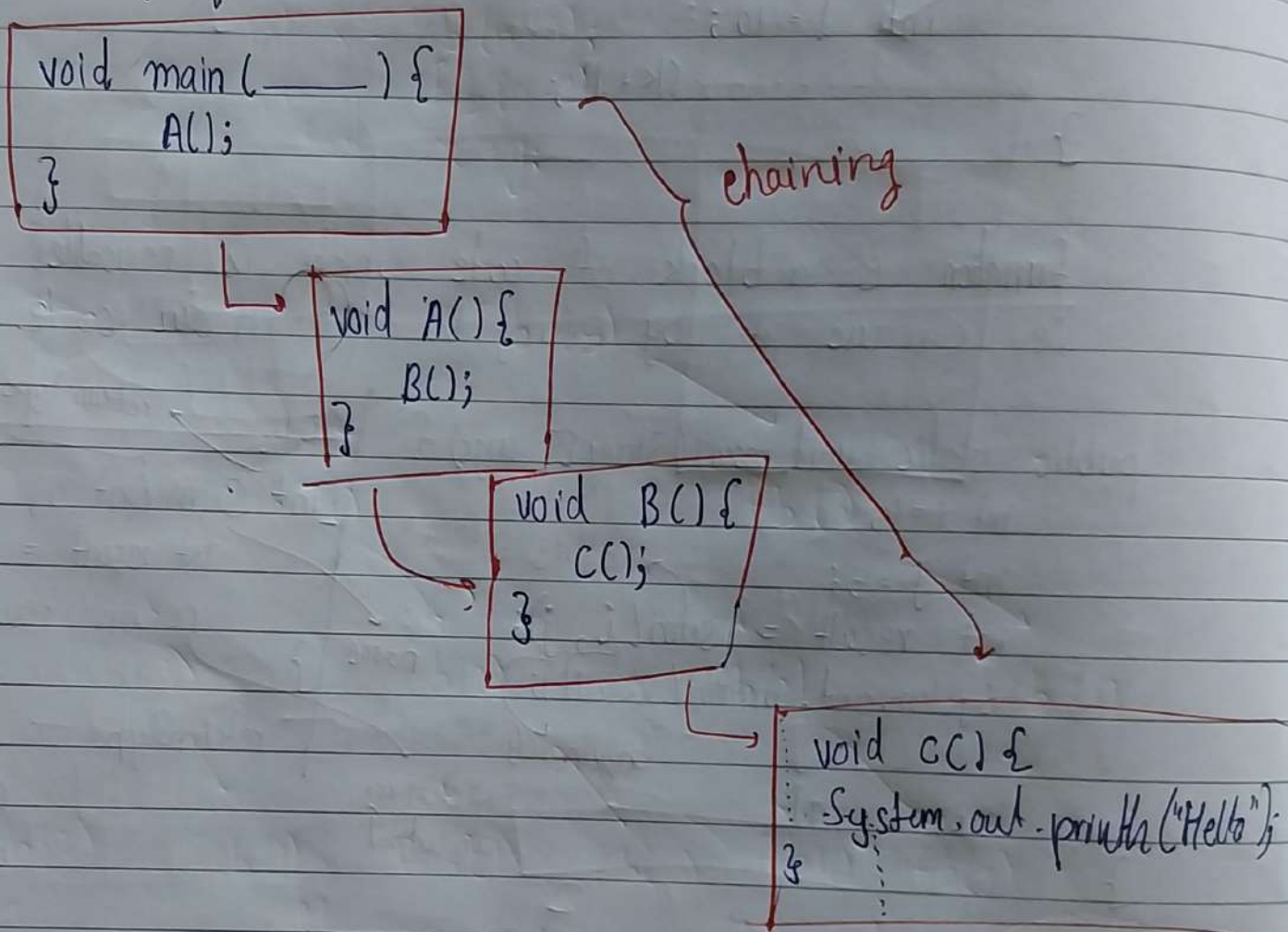
input $\longrightarrow$ functions $\longrightarrow$ output

* Types of functions

- No input, No output
- Input, No output
- No input, output
- input, output

if funcⁿ is not returning anything then we write void in place of datatype

* Chaining of functions



* Recursion

→ A function which call itself.

```
main( ) {
    A();
}

void A() {
    A();
}
```

Base case

print 1-10;

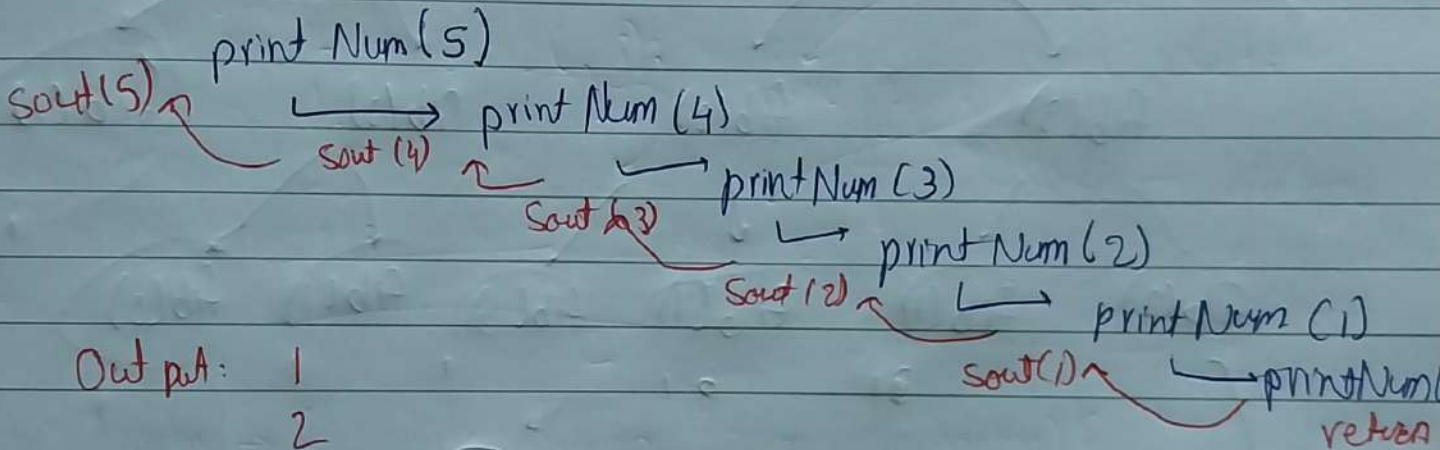
```
main() {
    print Num(5);
}
```

```
void printNum(int n) {
    if (n == 0) return;
    print Num(n-1); // Recursive call
    System.out.println(n);
}
```

Working

Shortcut

→ sout = System.out.println(n)



Out put:

1

2

3

4

5

```
void printNum(int n) {
    if (n == 0) {
        return;
    }
    printNum(n-1);
    sout(n);
}
```

→ Base case

printNum(n-1); → Recursive call

sout(n);

}

Using recursion to print fibonacci numbers;

fibonacci : ^{0th 1st 2nd 3rd 4th 5th}
1, 1, 2, 3, 5, 8, ...
series

n^{th} fibonacci no.

$$\text{Fib}(n) = \text{Fib}(n-1) + \text{Fib}(n-2)$$

```
int fib(n) { if (n == 0 || n == 1) return 1;  
             int x = fib(n-1);  
             int y = fib(n-2);  
             return x + y;  
}
```

